Journal of Automation and Intelligence xxx (xxxx) xxx

Contents lists available at ScienceDirect

Journal of Automation and Intelligence

journal homepage: www.keaipublishing.com/en/journals/journal-of-automation-and-intelligence/



Research article

Region-weighted gradient and motion-aware camera exposure control for robust visual odometry

Qirui Sun, Qi Ouyang *, Xiaolong Li, Zhaoying Cheng

School of Automation, Chongqing University, Chongqing, 401331, China

ARTICLE INFO

Keywords:
Computer vision for robotic application
Camera attributes control
Motion blur awareness
Visual odometry (VO)

ABSTRACT

Visual odometry (VO) often suffers from performance degradation in high dynamic range (HDR) scenarios due to uneven illumination conditions, adversely affecting localization accuracy. To address this challenge, we propose a camera exposure control algorithm designed to enhance the robustness of visual odometry by adaptively adjusting the camera's shutter speed and gain parameters to obtain high-quality images. Specifically, the algorithm employs a gradient-weighted sum of local image regions as a quality evaluation metric, and iteratively updates exposure parameters by computing the first-order derivative of this metric with respect to the exposure settings. Moreover, scene irradiance information is incorporated to determine an appropriate update step size. To further mitigate motion blur, we constrain the camera's maximum shutter time based on feedback from the visual odometry system. the efficacy of the proposed approach is validated through experiments conducted in both HDR and fast-motion scenarios. Additionally, we integrate the exposure control strategy into two representative VO frameworks, ORB SLAM3 and VINS MONO, and evaluate their performance in both indoor and outdoor environments. The experimental results demonstrate that our exposure control method significantly improves the overall accuracy and robustness of visual odometry systems.

1. Introduction

In recent years, visual SLAM algorithms have attracted extensive research attention, and a number of outstanding methods have been developed [1–4]. However, these algorithms face significant challenges under complex lighting conditions. In scenes characterized by drastic illumination changes, the luminance variation of captured images far exceeds the dynamic range of the sensor. Utilizing either fixed exposure or conventional auto-exposure strategies often results in overexposure or underexposure, thereby leading to the loss of visual feature points and erroneous feature matching. Furthermore, motion blur represents another critical challenge in such conditions; when the camera moves rapidly, an excessively long shutter time can cause severe motion blur, which is catastrophic for visual SLAM algorithms that require continuous tracking of feature points. Therefore, acquiring images with appropriate exposure parameters and minimized motion blur is a prerequisite for the robust operation of visual SLAM systems in HDR environments.

To address the challenges of imaging under complex illumination, existing research has explored multiple parallel directions. Common approaches include image enhancement [5–7], specialized imaging sensors [8,9], and camera exposure control [10,11]. The first two approaches can improve dynamic range or image details; However, learning-based or multi-frame fusion methods are computationally intensive and difficult to run in real time on VO systems [12], while specialized sensors introduce additional hardware and system integration costs. In contrast, exposure control requires no extra hardware, involves only lightweight real-time adjustments of camera parameters, and thus offers low computational overhead and strong portability, making it well-suited for direct integration into existing VO systems. Meanwhile, some studies have incorporated photometric adaptation within SLAM pipelines to enhance robustness under illumination changes [13], but such methods typically involve substantial modifications to the overall SLAM framework.

In this article, we propose an active exposure control method that can be seamlessly integrated into existing visual SLAM systems. As illustrated in Fig. 1, our exposure algorithm outputs examples of optimized images. The camera's exposure attributes are primarily determined by shutter time and gain. Shutter time is the duration during which the sensor captures light, and gain represents the amplification applied to the image signal; both parameters affect image brightness. A longer shutter time increases brightness but exacerbates motion blur, while

Peer review under responsibility of Chongqing University.
* Corresponding author.
E-mail address: yangqi@cqu.edu.cn (Q. Ouyang).

https://doi.org/10.1016/j.jai.2025.10.001
Received 6 May 2025; Received in revised form 19 September 2025; Accepted 13 October 2025

2949-8554/© 2026 The Authors. Published by Elsevier B.V. on behalf of KeAi Communications Co. Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

Please cite this article as: Q. Sun, Q. Ouyang, X. Li et al., Region-weighted gradient and motion-aware camera exposure control for robust visual odometry, Journal of Automation and Intelligence (2026), https://doi.org/10.1016/j.jai.2025.10.001.

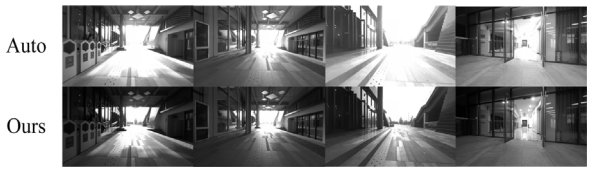


Fig. 1. Exposure control performance: Auto vs. Ours.

a higher gain also brightens the image but amplifies noise. Current exposure control algorithms typically depend on two components: (1) an image quality evaluation metric, and (2) an exposure update strategy.

We propose a dynamic camera exposure control method that adapts to local illumination variations and reduces motion blur, there by significantly improving the robustness of current VO algorithms in HDR scenarios. As depicted in Fig. 2, our approach utilizes the weighted sum of gradients in local image regions as an exposure evaluation metric and iteratively updates the exposure parameters by computing the first derivative of this metric with respect to the exposure settings, with the update step size determined adaptively base on scene irradiance.

Moreover, based on the displacement rate output from the VO algorithm, we calculate the maximum allowable shutter time for the current scene to further suppress motion blur. The corresponding gain is then derived based on this maximum shutter time, thereby defining the optimal camera exposure settings. The main innovations and contributions of this work are as follows:

I. We propose a camera exposure control method that accounts for local illumination changes and motion blur.

II. We introduce an exposure evaluation metric based on the gradient-weighted sum of image regions and develop a strategy for dynamically adjusting the exposure update step.

III. We validate our algorithm through extensive experiments conducted in diverse scenarios, using the output of VO algorithms to constrain shutter time in order to mitigate motion blur.

Section 2 reviews related work in image enhancement and camera exposure control. Section 3 describes the camera photometric response model as well as the image quality evaluation metric and its first-order derivative with respect to exposure. Section 4 details the exposure update strategy, including the allocation of shutter time and gain. Section 5 presents experimental validation of the proposed algorithm, and Section 6 concludes the paper.

2. Related work

Acquiring high-quality images is essential for ensuring the stable operation of visual simultaneous localization and mapping (SLAM) algorithms, as such images typically contain rich gradient information and clear visual details. To address imaging challenges under complex illumination conditions, several approaches have been proposed. First, image enhancement methods, including learning-based enhancement models and multi-frame high dynamic range (HDR) algorithms [5–7], can improve image details and dynamic range to a certain extent. Second, specialized imaging sensors, such as multi-gain/high-speed global shutter CMOS and event cameras, extend the dynamic range and effectively suppress distortions at the hardware level. [8] Third, camera exposure control methods directly improve image quality by adjusting parameters such as shutter time and gain in real time. This work mainly focuses on the application of exposure control in visual odometry (VO). Existing exposure control algorithms can generally be categorized into two types: learning-based methods and optimization-based methods. In addition, it is worth noting that recent studies have also proposed illumination-robust visual SLAM frameworks [13–15], which incorporate photometric adaptation mechanisms within the SLAM system itself to enhance stability under challenging illumination conditions.

2.1. Learning-based exposure control

Tomasi et al. [16] trained a lightweight network using matchable features and inliers as proxy metrics for predictive joint control of shutter and gain, improving VO stability under drastic illumination changes such as tunnel entry and exit. However, it requires dual-camera data collection, is tightly coupled with specific SLAM frontends, and lacks explicit shutter constraints, leaving motion blur unresolved in high-speed scenarios. Lee et al. [16] employed deep reinforcement learning with vectorized states and curriculum learning to achieve millisecond-level control and rapid convergence, but their reward was designed for general image quality rather than VO-specific objectives, and training relied on specialized hardware with limited cross-device generalization. He et al. [17] developed an offline simulator with bracketed exposure and photometric synthesis, and trained a SAC-based model with multi-level VO-aligned rewards, offering strong reproducibility. Yet, its performance is sensitive to photometric calibration and simulator fidelity, and cross-scene or cross-sensor deployment still requires additional adaptation.

2.2. Optimization-based exposure control

Optimization-based camera exposure control methods offer strong interpretability and generalization. They typically update exposure parameters iteratively to maximize image quality metrics, with image gradient and entropy being the most widely adopted indicators, both of which have proven effective across various computer vision tasks [18, 19]. Shim et al. [10,11] employed a weighted sum of gradient magnitudes as the quality metric and simulated gradient variations under different exposures using gamma correction, thereby determining the optimal exposure in outdoor scenes. Zhan et al. [20] introduced a soft-percentile gradient metric, derived the gradient of shutter time from the camera response function, and optimized exposure via gradient descent. Some studies incorporated image entropy as a complementary indicator: Kim et al. [21] combined entropy with gradient magnitude and applied Bayesian optimization to estimate optimal exposure; Lu et al. [22] utilized Shannon entropy to dynamically adjust shutter and gain along predefined trajectories; and Shin et al. [23] proposed a composite metric integrating gradient, entropy, and noise level, and employed the Nelder–Mead algorithm for fast convergence without gradient computation. A few works attempted to integrate optimization strategies directly into visual SLAM. For example, Han et al. [12] proposed a framework that maximized a gradient-entropy metric via gradient descent while estimating scene dynamics through the optical flow of central pixels between consecutive frames to alleviate motion blur. However, this approach is prone to erroneous evaluations in fast straight-line motion where optical flow remains minimal, and it requires additional optical flow modules for feature-based VO systems, which increases computational overhead and limits general applicability.

Despite the notable progress of optimization-based approaches, existing methods still exhibit certain limitations. Most rely on global gradient or entropy metrics, which are insensitive to local under- or over-exposure and may induce exposure jitter. Moreover, many approaches [10,11,20,21] fail to explicitly constrain shutter time with respect to scene dynamics, or they depend on optical flow [12] for estimating change rates, which is error-prone in specific scenarios and lacks universality for VO frameworks that do not employ optical flow.

3. Region-Weighted Gradient (RWG) metric

Gradient information of images is extensively applied in various computer vision tasks, such as feature extraction [24] and edge detection [25]. Existing studies commonly employ the sum of gradient magnitudes as an indicator for assessing image quality but fail to consider overexposed regions within images. Although overexposed

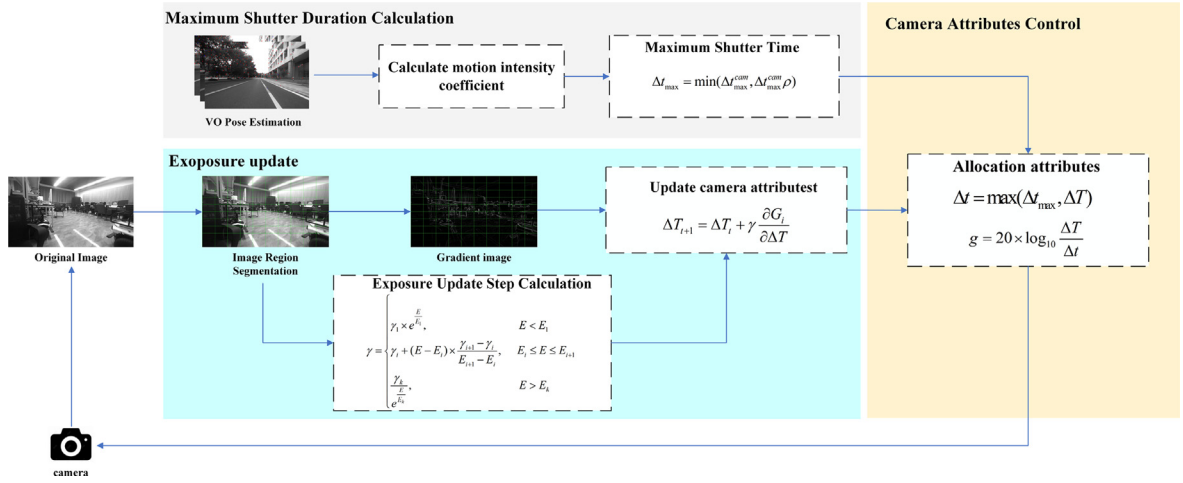


Fig. 2. Pipeline of the proposed method.

areas can increase the overall gradient values, they actually exacerbate overexposure, causing loss of local image details and consequently reducing overall image quality. To address this issue, we utilize a region-based segmentation method to filter out overexposed areas, assigning them lower weights. Although this might slightly decrease the overall gradient information, it preserves more detailed information, thus providing a more accurate representation of the visual features of real scenes.

3.1. Photometric response function

In this paper, we compute the camera's photometric response model according to [26], and the imaging process is illustrated in Fig. 4 (a). Where E denotes the irradiance of the scene, representing the energy received per pixel per unit time; X is the total accumulated energy received by each pixel over the shutter duration; and g denotes the gain, defined as the amplification factor for the analog image signal, calculated according to (1). The gain and shutter time jointly determine the captured image quality, with the received energy being transformed into pixel intensity through a nonlinear mapping function f. Here ε is electronic gain (amplification factor, dimensionless) and g is the linear gain.

$$\varepsilon = 10^{\frac{g}{20}} \tag{1}$$

$$I = f(X) = f(E\varepsilon\Delta t) \tag{2}$$

The function f is invertible, allowing us to define the inverse camera response function as follows:

$$z(I) = \ln f^{-1}(X) = \ln E + \ln \varepsilon \Delta t \tag{3}$$

Here, E denotes the irradiance of the scene at the sensor surface, Δt is the shutter time (exposure duration), ε is the sensor electronic gain (amplification factor, dimensionless), X represents the accumulated energy during exposure, and $f(\cdot)$ is the camera response function, which is assumed to be monotonic and invertible.

As shown in Fig. 3, similar image appearances can be achieved through various combinations of exposure time and gain. Based on this equivalence, we define a generalized camera exposure ΔT as the product of shutter time and analog signal amplification factor:

$$\Delta T = \Delta t \times \varepsilon \tag{4}$$

Using the inverse camera response function, we can recover the scene irradiance from pixel values. By capturing a series of images at different exposures and gains in a static scene and performing photometric calibration as described in [26], we obtain the calibration sequence and inverse response function illustrated in Fig. 4 (c). These discrete values are fitted using a 10th-order polynomial, from which the derivative z' is subsequently calculated.

3.2. Gradient metrics

Gradient intensity is an essential metric for evaluating image quality. For feature-based visual odometry, such as ORB-SLAM [3], image gradients determine the quality and quantity of extracted features, thus affecting the accuracy of pose estimation. In optical flow-based visual odometry, like VINS-MONO [2], features are first extracted from the image, followed by feature tracking based on pixel intensity values, and finally matched features are used to solve for poses. Therefore, acquiring images rich in gradient information is a prerequisite for stable feature tracking in these visual SLAM algorithms.

However, camera property adjustments are global, making it difficult to cope with local illumination changes. When parts of the image are severely overexposed, while other regions are normal or underexposed, global metrics become averaged, adversely affecting normal regions. Particularly for optical flow-based odometry, local illumination changes may cause the camera to reduce gain or exposure to enhance overall gradients, inadvertently violating the illumination invariance assumption and thus failing optical flow tracking, leading to decreased odometry accuracy.

To address this, we employ region-based segmentation to handle local overexposure or underexposure. We formalize the region-weighted gradient metric as follows. The image is divided into k rectangular regions of roughly equal size $\{R_i\}_{i=1}^{k}$. For each pixel u in region R_i, the gradient magnitude is computed as

$$\tilde{g}(u) = \|\nabla I(u)\|_2^2 = \left(\frac{\partial I(u)}{\partial x}\right)^2 + \left(\frac{\partial I(u)}{\partial y}\right)^2 \tag{5}$$

where $\frac{\partial I(u)}{\partial x}$ and $\frac{\partial I(u)}{\partial y}$ denote horizontal and vertical gradients obtained with a Sobel operator. The regional gradient score is computed by aggregating the pixel-level gradients within the region. To account for overexposure, we count the number of pixels whose intensity values exceed the saturation threshold; these pixels are regarded as severely overexposed. The gradient weight of the region is then determined according to the proportion of overexposed pixels relative to the total number of pixels in the region.

$$G_i = \frac{1}{|R_i|} \sum_{u \in R_i} \tilde{g}(u) \tag{6}$$

$$s_i = \frac{1}{|R_i|} \sum_{u \in R_i} \mathbb{I}(I(u) > I_{\text{sat}}), \qquad w_i = e^{-\alpha s_i} \quad \alpha > 0 \tag{7}$$

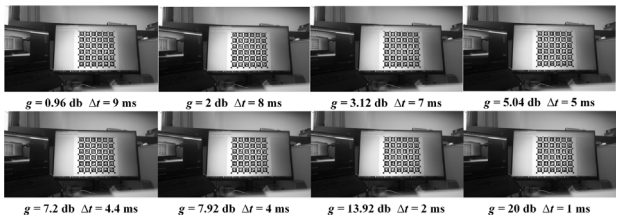


Fig. 3. Equivalence of Shutter Time and Gain.

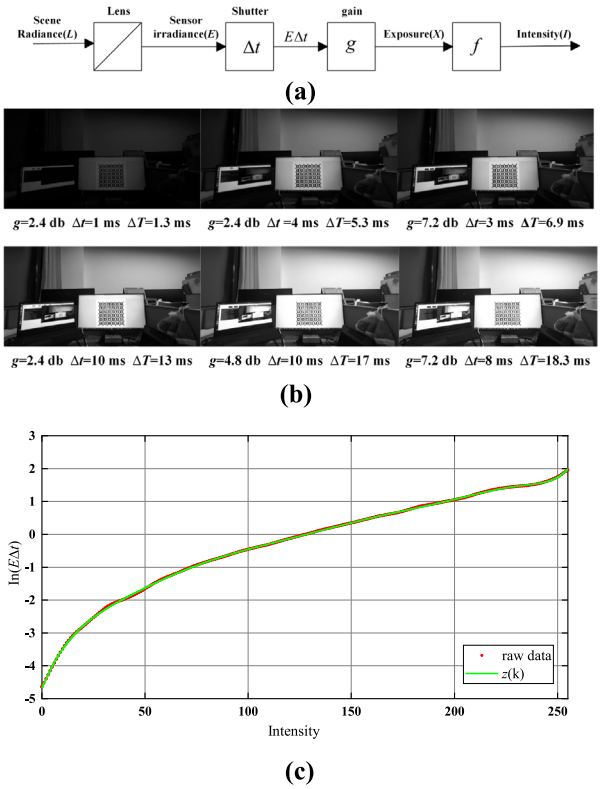


Fig. 4. Camera Imaging Process and Photometric Response Function.

Where $|R_i|$ is the number of pixels in region. s_i represents the proportion of overexposed pixels in the region R_i, and w_i is the weight of the region calculated based on s_i.

So that heavily saturated regions receive lower weights. Finally, the global image quality score is computed as the weighted average across all regions:

$$G = \frac{\sum_{i=1}^{k} w_i G_i}{\sum_{i=1}^{k} w_i} \tag{8}$$

Here, G reflects the overall sharpness and contrast of the image while suppressing the influence of locally overexposed regions.

Following Shin et al. [23], we derive the first-order derivative of the image quality metric G with respect to the generalized exposure ΔT. The pixel intensity is given by Eq. (2). We assume $f(\cdot)$ is monotonic, invertible, and differentiable. Let the log inverse response be $z = \ln f^{-1}$. From $z(I) = \ln E + \ln \Delta T$ one obtains, by the chain rule

$$\frac{\partial I(\mathbf{u})}{\partial \Delta T} = \frac{1}{z'(I(\mathbf{u}))\Delta T} \tag{9}$$

With $z'(I)$ computed analytically from the calibrated inverse response $z(I)$. Given an image, denoted as I, captured with an exposure time ΔT, the magnitude of the gradient at a pixel u is

$$\tilde{g}(\mathbf{u}, \Delta T) = \|\nabla I(\mathbf{u}, \Delta T)\|_2^2 = \left(\frac{\partial I(u, \Delta T)}{\partial x}\right)^2 + \left(\frac{\partial I(u, \Delta T)}{\partial y}\right)^2 \tag{10}$$

Using Schwarz's theorem $\frac{\partial}{\partial \Delta T}\nabla I = \nabla\left(\frac{\partial I}{\partial \Delta T}\right)$, calculate the derivative of the pixel value of pixel u in image I with respect to exposure.

$$\frac{\partial I(u, \Delta T)}{\partial \Delta T} = f'(f^{-1}(I))E(u) = \frac{E(u)}{[f^{-1}(I)]'} = \frac{1}{z'(I)\Delta T} \tag{11}$$

$$\begin{aligned} \frac{\partial \tilde{g}(\mathbf{u}, \Delta T)}{\partial \Delta T} &= 2\nabla I(\mathbf{u}, \Delta T)^{\top} \frac{\partial \nabla I(\mathbf{u}, \Delta T)}{\partial \Delta T} \\ &= 2\nabla I(\mathbf{u}, \Delta T)^{\top} \nabla\left(\frac{1}{z'(I)\Delta T}\right) \end{aligned} \tag{12}$$

where E represents the scene irradiance received by pixel u, and $z(I)$ is the logarithmic inverse response function of the camera, which is obtained by polynomial fitting, $z'(I)$ is the first derivative of $z(I)$. Subsequently, the pixel-wise derivatives are accumulated over all pixels within a region to obtain the derivative of the regional gradient sum with respect to ΔT.

$$\frac{\partial G_i}{\partial \Delta T} = \frac{1}{|R_i|}\sum_{\mathbf{u}\in R_i} \frac{\partial \tilde{g}(\mathbf{u})}{\partial \Delta T}, \quad \frac{\partial G}{\partial \Delta T} = \frac{\sum_{i=1}^{k} w_i \frac{\partial G_i}{\partial \Delta T}}{\sum_{i=1}^{k} w_i}. \tag{13}$$

Where R_i is region i and w_i is its exposure-aware weight. In practice, w_i is treated as piecewise constant with respect to ΔT (it only changes when pixels cross the saturation threshold); therefore we recompute w_i per iteration and omit $\partial w_i/\partial \Delta$T except at threshold events.

4. Camera attributers control

The camera parameter control method consists of two primary steps. First, exposure values are updated using gradient descent based on the predicted derivative, with the step size jointly determined by the derivative and the scene irradiance. Second, gain and shutter time are allocated based on the calculated exposure.

4.1. Exposure control scheme

Specifically, gradient descent is utilized to update exposure, adjusting the exposure along the negative gradient direction to maximize the weighted sum of image gradients for each frame.

$$\Delta T_{t+1} = \Delta T_t + \gamma \frac{\partial G}{\partial \Delta T} \tag{14}$$

The gradient descent step size γ depends on the scene irradiance E. We sample irradiance values E_i ($i = 1, 2, 3, \ldots,$ k) from$k$ scenarios, each assigned a fixed step size $\gamma_i (i = 1, 2, 3, \ldots,$ k). The sample irradiance gradually transitions from low to high levels. In low irradiance conditions, a larger step size is desired to quickly enhance brightness and a smaller step size is preferred in high irradiance conditions to prevent overexposure. Hence, for irradiance levels below the lowest sample, the gradient step size increases exponentially, while for levels above the highest sample, it decreases exponentially; within the sampled range, linear interpolation determines the step size.

$$\gamma = \begin{cases} \gamma_1 \times e^{\frac{E}{E_1}}, & E < E_1 \\ \gamma_i + \left(E - E_i\right) \times \frac{\gamma_{i+1} - \gamma_i}{E_{i+1} - E_i}, & E_i \le E \le E_{i+1} \\ \frac{\gamma_k}{e^{\frac{E}{E_k}}}, & E > E_K \end{cases} \tag{15}$$

Where γ is the gradient update step size for the current irradiance E, with γ_1 corresponding to the lowest irradiance E_1, and γ_k corresponding to the highest irradiance E_K. This dynamic strategy rapidly enhances image gradients in low-light scenarios while gradually adjusting exposure under high-light conditions to avoid overexposure.

4.2. Shutter time or gain

Both shutter time and gain influence image brightness. However, longer shutter times can introduce motion blur in high-speed scenes, and higher gain levels amplify image noise, reducing overall image quality. Therefore, after calculating the optimal exposure, it is necessary to appropriately allocate shutter time and gain to achieve superior image quality. Our method considers the camera's motion scenario, prioritizing increased shutter time in static or low-speed scenes to enhance image quality, and favoring increased gain in high-speed scenarios to avoid motion blur.

The visual odometry output position rate is used as a metric for scene dynamics. When the positional update is rapid, the camera moves quickly. For VIO algorithms, IMU data directly provide motion information. The frontend of VO or VIO computes the relative translational change ΔP_i and the relative rotational change ΔR_i between two consecutive frames. The rotational vector norm is then obtained using Eq. (16), and together with the pose variation, the maximum exposure time is calculated using Eq. (17). This guarantees that the shutter duration remains stable under the combined constraints of translation and rotation:

$$\| \Delta\theta_k \| = \arccos\left(\frac{\mathrm{tr}(\Delta R_k) - 1}{2}\right) \tag{16}$$

$$\rho = \frac{\lambda v_{ref} \Delta\tau_i}{\|\Delta P_i\|} + \frac{\eta w_{ref} \Delta\tau_i}{\|\Delta\theta_i\|} \tag{17}$$

$$\Delta t_{\max} = \min(\Delta t_{\max}^{cam}, \Delta t_{\max}^{cam} \rho) \tag{18}$$

Where v_{ref} and w_{ref} denote the reference linear and angular velocities, respectively, used for nondimensionalization. $\Delta\tau_i$ denote the time interval between consecutive odometry pose updates. Parameters λ and η are obtained through experimental calibration, representing the sensitivities to translation and rotation. The final camera settings are computed by (19) and (20), Δt and g represent the shutter time and gain for the next frame.

$$\Delta t = \max(\Delta t_{\max}, \Delta T) \tag{19}$$

$$g = 20 \log \frac{\Delta T}{\Delta t} \tag{20}$$

Shutter time and gain similarly enhance brightness, and their combined effect is the product of the two. Extended shutter times result in motion blur, while higher gains introduce noise. Motion blur impacts visual odometry algorithms more severely than noise by significantly compromising the accuracy of feature matching or optical flow tracking, potentially causing tracking failures. Thus, we adopt a shutter-priority exposure strategy, calculating the maximum shutter time without blur based on the current motion scenario, and compensating any excess exposure through gain adjustments.

The detailed procedure is summarized in Algorithm 1. First, the current camera exposure is computed using Eq. (4), and the input image is uniformly divided into K regions to calculate the proportion of overexposed pixels in each region. The regional weights are then determined according to Eq. (7), and the weighted sum of regional gradients yields the image quality metric G. Based on Eq. (15), the first-order derivative of the quality metric with respect to exposure is further obtained. Considering the irradiance of the current scene, the gradient update step size is calculated using Eq. (19), which is then used to update the current exposure and determine the optimal exposure value. Meanwhile, the scene variation rate is estimated from the output of the VO algorithm, from which the maximum shutter time is derived for the current scenario. Finally, exposure is allocated according to this maximum shutter time, resulting in the camera exposure parameters for the current image frame.

Table 1
Camera parameters.

Camera Model	Image size	Shutter
MV-EM200M	640×480	Global

Algorithm 1: Camera Exposure Control with Motion Blur and Local Lighting Aware

Input: Current image I_i, VO position output P_i, current shutter time Δt_i, current gain g_i
Output: Set shutter time Δt_{i+1} and gain g_{i+1}

1 Calculate current exposure ΔT_i by Eq. (4)
2 Partition image into K regions and compute weight w_i for each region

$$w_i = \exp(-\lambda \frac{1}{n} \sum_{u \in R_i} \mathbb{I}(I(\mathbf{u}) > I_{\text{sat}}))$$

3 Compute weighted sum of regional image gradients G_i using Eq. (8)
4 Calculate derivative of weighted gradient sum exposure $\frac{\partial G}{\partial \Delta T}$ using Eq. (13)
5 Compute exposure update step size γ by Eq.(15)
6 Update exposure parameters: $\Delta T_{t+1} = \Delta T_t + \gamma \frac{\partial G}{\partial \Delta T}$
7 Calculate maximum allowable shutter time Δt_{max} using Eq. (18)
8 Update shutter time Δt_{i+1} and gain g_{i+1}
9 **return** Δt_{i+1}, g_{i+1}

5. Experiments

In this section, we first describe our experimental setup and then systematically evaluate the proposed image-quality evaluation metrics against existing methods across various scenarios to demonstrate both accuracy and robustness. To verify the effectiveness of our algorithm in suppressing motion blur, we mounted the camera on a mobile robot moving at a constant speed, thereby demonstrating our algorithm's rapid adaptability in practical visual odometry (VO) operations. To test the convergence speed under significant lighting variations, we conducted experiments involving rapid switching of indoor lighting conditions. Additionally, we integrated our method into both ORB SLAM3 and VINS MONO frameworks, conducting VO tests under diverse conditions to further confirm the robustness enhancement provided by our exposure-control algorithm. Finally, we measured computational latency to verify the real-time feasibility of our method with contemporary VO systems. In all experiments, we set i = 400, α = 0.4, λ = 0.17, η = 0.12, v_{ref} = 5 m/s, w_{ref} =2 rad/s, γ ={2.5, 2.0, 1.3, 0.3, 0.002 0.0008 0.0003}, E ={0.00006, 0.000083, 0.000110, 0.000240, 0.001 0.00266, 0.008}, all of which were empirically determined.

5.1. Experimental setup

Fig. 5 illustrates our experimental hardware configuration. We employed a camera module comprising two identical cameras capturing the same scene simultaneously, each controlled by different exposure algorithms, to directly compare their effects on image quality. The camera parameters are detailed in Table 1. The camera system is mounted on a four-wheeled mobile robot, which is also equipped with LiDAR, an inertial measurement unit (IMU), and real-time kinematic GPS (RTK-GPS). Indoor scenarios leveraged high-accuracy laser-inertial odometry LIO-SAM [26] as ground truth, while outdoor scenarios used RTK-GPS for evaluating VO accuracy.

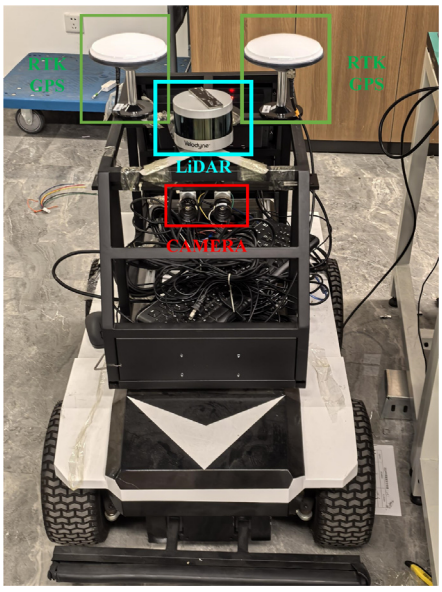


Fig. 5. Experimental Equipment. The mobile robot is equipped with a camera, LiDAR, and RTK-GPS.

5.2. Validation of control algorithms

The primary aim of this study is to validate the effectiveness of our proposed exposure-control algorithm within visual SLAM contexts. We conducted systematic real-world experiments encompassing diverse lighting conditions. Quantitative image-quality evaluation involved the number of ORB feature points extracted per frame. Given that current mainstream visual SLAM methods rely heavily on this metrics, our evaluations effectively highlight the algorithm's capability to enhance overall SLAM performance. In the experimental section, we compared the number of feature points extracted by the proposed algorithm with those obtained using the methods of Zhang et al. [20] and He et al. [17]. Specifically, Zhang's method calculates shutter time based on gradient percentiles and is implemented in C++, while He's method adopts a data-driven strategy. Following the description in their paper, we collected 3000 images across multiple scenes under five exposure levels and trained the model offline in a simulator; the corresponding code is publicly available and implemented in Python.

Fig. 6 illustrates the comparison of ORB feature point counts over 3000 frames (with the maximum number of features per frame set to 4000). Under stable illumination conditions (frames 0–1000), the performance of the three methods shows no significant differences, with He's method achieving overall better results. However, in the low-light region (Region A), He's method performs slightly worse than ours due to insufficient coverage of dark samples in its training dataset. In the corridor region (Region B), where local overexposure occurs, all methods experience a rapid decline in feature point counts. Nevertheless, our method mitigates the effect of local overexposure by introducing region-based weighting, thereby outperforming the compared methods. After Region B, the illumination drops sharply below the hardware's exposure limit, under which all algorithms exhibit comparable performance. Overall, He's method demonstrates certain advantages under normal illumination; however, in low-light environments and scenarios with uneven illumination, our method exhibits stronger robustness. In contrast, Zhang's method lacks effective gain control, increasing gain only after the shutter time reaches its upper limit, and its fixed update step size results in weaker responsiveness under scenes with drastic illumination changes.

5.3. Validation of extreme illumination

To quantitatively assess the response speed of our algorithm under drastic lighting transitions, we conducted an indoor experiment involving rapid toggling of artificial lighting. Initially placing the camera in a dim environment, sudden illumination triggered immediate overexposure, gradually adjusted to optimal exposure by our algorithm. Conversely, switching off the light instantly resulted in image underexposure, subsequently corrected by dynamically increasing exposure. This procedure effectively constituted applying two sequential step signals to the camera, allowing us to evaluate transient responses. Experimental results are shown in Fig. 7. which clearly indicate superior performance of our algorithm over the camera's built-in exposure control, as our method achieved high-quality exposure corrections within 200 ms, significantly faster than the built-in method's 300 ms adjustment period.

Furthermore, we compared the response speed of the exposure control algorithm with and without the dynamic step-size mechanism, with the fixed step size set to $\gamma =$ (to be specified). The experimental results are shown in Fig. 8(b). When the light was turned on, the algorithm with the dynamic step-size mechanism reached a stable state within approximately 200 ms, whereas the fixed-step algorithm required about 533 ms to converge. In the light-off scenario, the dynamic step size algorithm stabilized within about 100 ms, while the fixed-step algorithm still required considerably more time. Overall, the dynamic step-size mechanism accelerated the convergence speed by nearly fourfold compared to the fixed-step approach, significantly enhancing the algorithm's responsiveness under abrupt illumination changes.

5.4. Validation of motion blur suppression

To further quantitatively evaluate the effectiveness of motion-blur suppression, we designed the experiment illustrated in Fig. 9. In this test, the camera was translated linearly at a speed of 2 m/s and rotated about the z-axis at an angular velocity of 4 rad/s. Data collection was restricted to periods of constant velocity to eliminate the effects of acceleration and deceleration. For comparison, He's method did not provide a rule for shutter and gain allocation; thus, we defined the shutter time to be inversely proportional to both translational and rotational velocity. Zhang's method requires a predefined maximum exposure time, which was set to 15 ms in this experiment. The results demonstrate that our method reduces gain in static scenarios to improve image quality, while in high-speed motion scenarios it rapidly decreases shutter time to effectively mitigate motion blur. Compared with He's method, Zhang's method, and the built-in automatic exposure algorithm, our approach shows a significant advantage in motion-blur suppression, thereby substantially improving image quality under dynamic operating conditions.

In addition, we compared the performance of the algorithm with and without the motion-blur suppression module. As shown in the last column of Fig. 9, the algorithm without the suppression module produced images with pronounced blur. Fig. 8(a) further presents the comparison of shutter time between the two methods, indicating that the algorithm without the suppression module employed significantly longer shutter times than our proposed control strategy, and thus lacked effective suppression of motion blur.

5.5. Camera attribute control for visual odometry

In the final step, we validated the performance of the proposed camera control algorithm in real-world scenarios using visual SLAM systems. A total of three groups of experiments were conducted. First, we verified the advantages of our exposure control algorithm over the built-in automatic exposure strategy in improving visual odometry accuracy under indoor and outdoor low-speed motion scenarios.

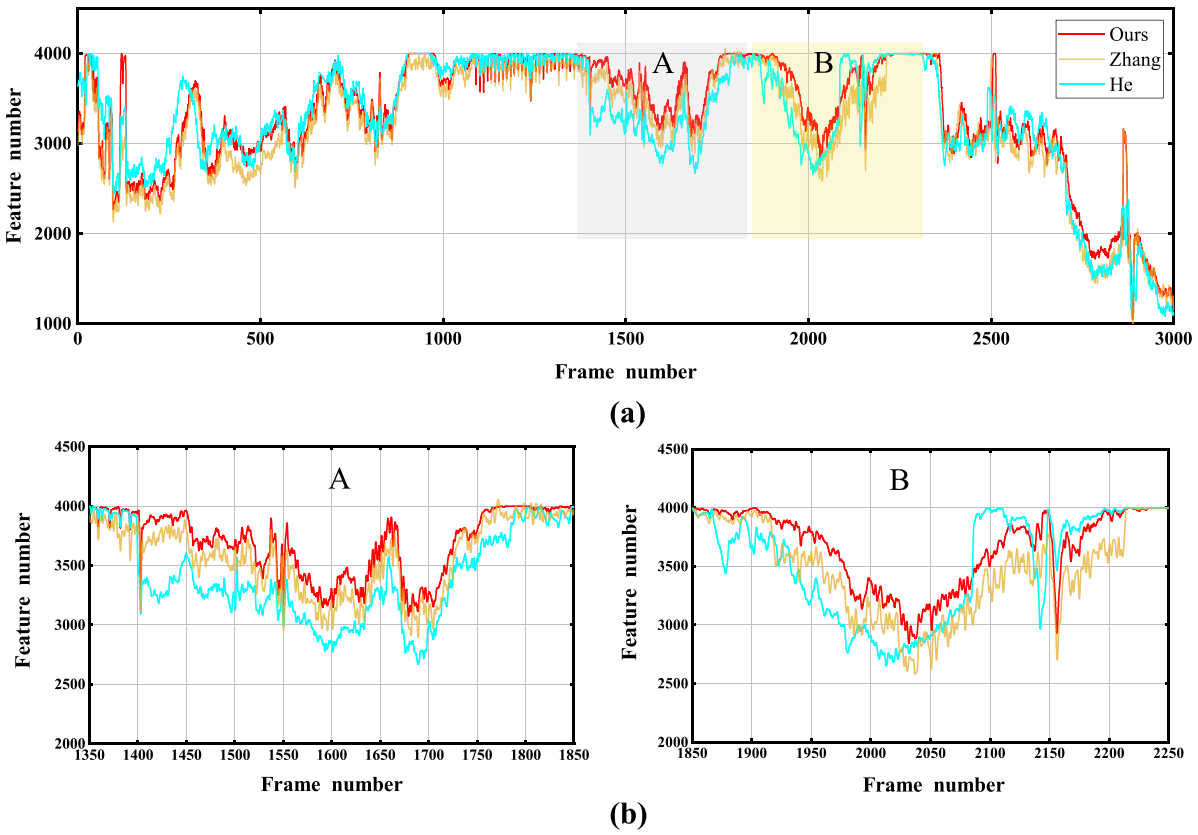


Fig. 6. Comparison of extracted feature point numbers under different exposure control algorithms.

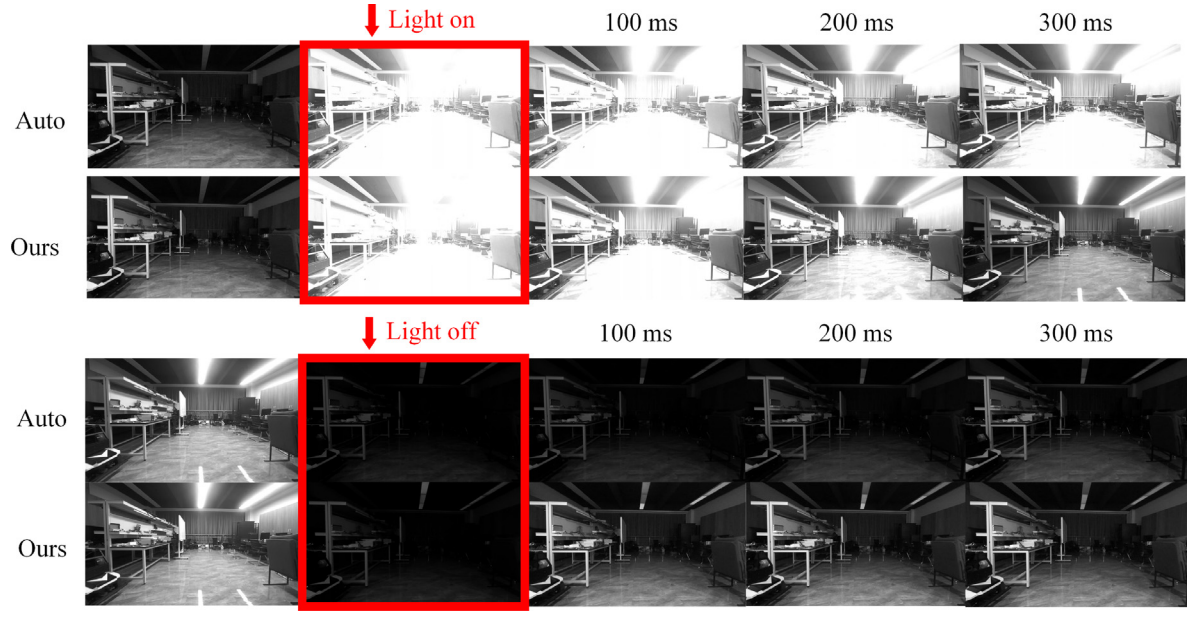


Fig. 7. Comparison of exposure adjustment speed under sudden light on/off events.

Subsequently, we compared our method with the latest learning-based algorithm in high-speed motion scenarios to further evaluate its performance under fast-motion conditions. A module consisting of two identical cameras, as shown in Fig. 5, was mounted on a mobile robot, with each camera running a different exposure control algorithm. To evaluate the algorithm performance, the image sequences collected by both cameras were input into ORB-SLAM3 and VINS-MONO for assessment. To eliminate uncertainties, the median of the trajectories obtained over ten runs of each algorithm was used as the result for the figures.

ORB-SLAM3 is a feature-based visual odometry algorithm that extracts ORB features from images and matches them between consecutive frames to estimate the camera pose. VINS-MONO is a monocular visual odometry system integrated with an Inertial Measurement Unit (IMU), which uses monocular vision to extract features and tracks them using optical flow. It then matches the features between consecutive

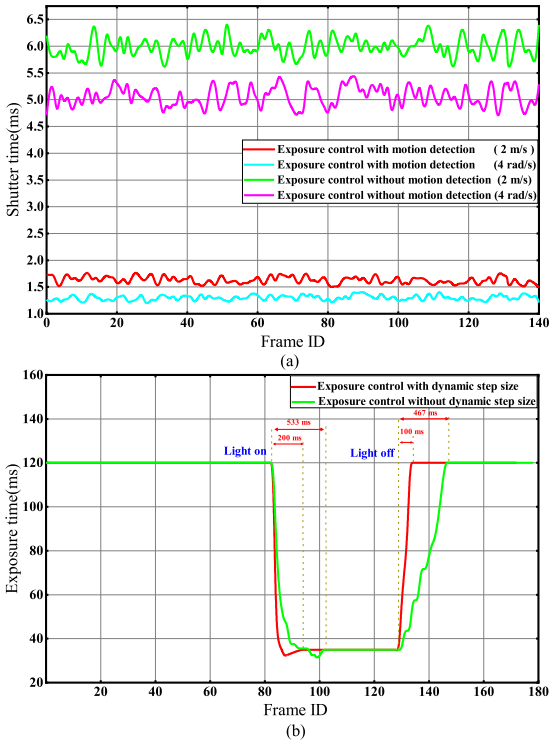


Fig. 8. Motion suppression and adaptive step-size analysis. (a) Shutter time variations with and without the motion-suppression module. (b) Exposure adjustment with and without the dynamic step-size module.

frames and combines the IMU information to estimate the camera pose. ORB-SLAM3 is sensitive to lighting changes, the number of features fluctuates significantly under rapid lighting variations, leading to feature matching failures and triggering the relocalization module. VINS-MONO, while benefiting from IMU information to compensate for visual deficiencies, suffers from increased reprojection error when the exposure is inappropriate in sudden lighting changes, impacting the system's state estimation accuracy.

5.5.1. Accuracy evaluation in low-speed scenarios

In the indoor scenario, the trajectories estimated by VINS-MONO and ORB-SLAM3 are shown in Fig. 10(a). The indoor scene passes through areas with low light and rapid lighting variations, causing the image quality to degrade quickly. For VINS-MONO, the IMU compensates when the image quality is poor. However, when running VINS-MONO with a camera equipped with the automatic exposure control algorithm, the trajectory error in the low-light areas is significant. Without loop closure constraints during the process, the accumulated error increases, causing the overall trajectory to drift noticeably. In contrast, our control algorithm responds quickly to environmental changes, resulting in smaller trajectory errors when using the camera with our algorithm compared to the automatic exposure algorithm. ORB-SLAM3, which lacks IMU constraints and has a high requirement for image quality, shows significant trajectory deviation when the image quality deteriorates severely. In low-light and rapidly changing lighting areas, both cameras equipped with automatic exposure algorithms show notable deviations, but the trajectory obtained with our exposure control algorithm is closer to the ground truth.

In the outdoor scene, the trajectories estimated by VINS-MONO and ORB-SLAM3 are shown in Fig. 10(b). The outdoor scene includes areas with rapid lighting changes, and our exposure control algorithm can quickly adjust exposure parameters in HDR scenarios. As a result, both VINS-MONO and ORB-SLAM3 show more accurate results when using our exposure control algorithm compared to the automatic exposure algorithm.

Table 2
Visual odometry accuracy in indoor scenarios.

Indoor	(Fig. 10(a))	ATE (m)	RPE_trans (m)	RPE_rot (°)
Auto	VINS	2.83	0.54	1.64
	ORB3	4.57	1.32	2.48
Ours	VINS	2.47	0.43	1.48
	ORB3	3.84	0.94	2.31

Table 3
Visual odometry accuracy in outdoor scenarios.

Outdoor	(Fig. 10(b))	ATE (m)	RPE_trans (m)	RPE_rot (°)
Auto	VINS	6.48	1.42	3.14
	ORB3	8.37	2.63	4.27
Ours	VINS	4.56	0.93	2.43
	ORB3	5.42	2.17	2.97

5.5.2. Accuracy evaluation in high-speed scenarios

In addition, we evaluated the visual odometry (VO) accuracy of the proposed algorithm under both low-speed and high-speed motion scenarios. Fig. 10(c) presents the trajectory outputs of the VINS-MONO algorithm equipped with three different exposure control methods. Since the monocular ORB-SLAM3 algorithm tends to crash and enter relocalization in high-speed scenarios, the comparison was limited to VINS-MONO.

In the low-speed scenario, both the proposed method and He's method significantly outperformed the automatic exposure algorithm. At turning points with drastic illumination changes, the automatic exposure algorithm failed to adapt effectively, resulting in pronounced pose drift and accumulation of errors. As He's method directly incorporates feature tracking into its training process as an evaluation metric, it exhibited stronger feature-tracking capability than our approach in this case.

In the high-speed scenario, the test vehicle traveled at approximately 20 km/h along four straight segments. Since He's method does not provide a rule for shutter and gain allocation, we manually set the maximum shutter time to 10 ms and compensated for the remaining exposure with gain. Experimental results show that all three algorithms experienced varying degrees of pose drift in high-speed conditions, particularly during sharp turns, where rapid camera rotation caused drastic scene changes and significantly degraded feature tracking performance. Nevertheless, the proposed method effectively suppressed both translational and rotational motion blur, improving the feature matching success rate and achieving superior performance compared with the other two methods under fast-motion conditions.

Tables 2 and 3 report the root mean square error (RMSE) of absolute trajectory error (ATE) and relative pose error (RPE) when running VO with the automatic exposure algorithm and the proposed method. Table 4 further summarizes the ATE and RPE results of VINS-MONO with three exposure control strategies under both low- and high-speed scenarios. The results indicate that the proposed algorithm consistently outperforms the automatic exposure strategy in both scenarios; although He's data-driven method shows slightly better performance in well-lit, low-motion environments, our approach achieves higher accuracy in high-speed conditions by constraining the shutter time using VO outputs, thereby effectively mitigating motion blur.

5.6. Processing time analysis

To verify whether the proposed algorithm can be integrated into existing VO or visual SLAM systems without imposing additional computational overhead, we conducted a detailed runtime analysis of its individual modules. The algorithm was implemented in C++ and tested on an i5-8300H CPU @ 2.30 GHz processor. We measured the execution time under different numbers of region partitions, including the computation time of the image quality metric, the time required to calculate

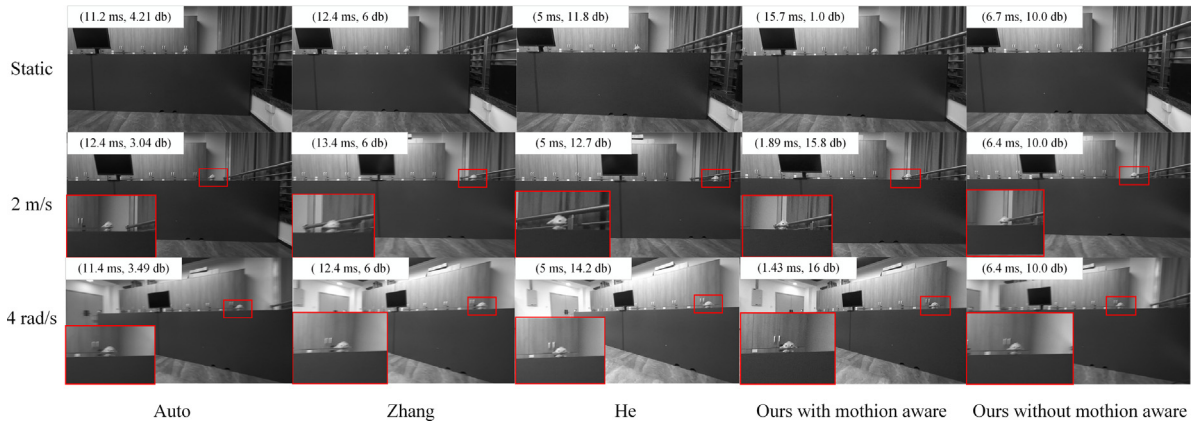


Fig. 9. Comparison of motion blur suppression under different exposure control algorithms.

Table 4
Visual odometry accuracy in slow and fast motion scenarios.

Slow&Fast	(Fig. 10(c))	ATE (m)	RPE_trans (m)	RPE_rot (°)
Auto	VINS (fast)	9.56	4.48	3.42
	VINS (slow)	4.32	0.60	2.87
Ours	VINS (fast)	6.46	1.87	2.87
	VINS (slow)	3.18	0.48	2.57
He	VINS (fast)	7.84	2.48	2.88
	VINS (slow)	2.12	0.46	2.29

the first-order derivative of the metric with respect to exposure, and the total processing time per frame. The implementation employed the parallel processing API of OpenCV, and the total frame processing time was consistently maintained within 10 ms. Upon receiving each image, the system immediately updated the exposure parameters; in static scenarios, convergence to a stable state was achieved within approximately three iterations, while in dynamic scenarios, the exposure parameters adapted promptly to scene changes. The results demonstrate that the proposed method completes all computations within 10 ms per frame, ensuring high real-time performance without placing significant computational demands on existing VO or SLAM systems (see Table 5).

6. Discussion

In this study, we proposed an image quality evaluation metric based on the region-wise gradient-weighted sum, which effectively reduces the influence of local overexposure on overall image quality through regional segmentation and overexposure ratio weighting. However, the weighting scheme mainly targets overexposed pixels and lacks a mechanism to address underexposed pixels. In addition, the robustness of the algorithm can be affected by the selection of segmentation granularity and pixel thresholds. Moreover, the method relies on the camera's photometric response function, which varies across different cameras, thus requiring additional photometric calibration when porting the algorithm.

Furthermore, the algorithm contains several empirically determined parameters (e.g., λ and η in Eq. (17)), which are not generalizable due to differences in CMOS sensor characteristics among cameras. These parameters need to be redefined in new experimental environments, thereby increasing the difficulty of algorithm migration. In the motion-awareness module, our method relies on the output of VO algorithms to estimate scene change rates. If the VO output undergoes large drifts, the scene change rate may be inaccurately estimated, leading to an excessively compressed maximum shutter time and consequently a rapid degradation in image quality.

7. Conclusion

In this work, we presents an image-quality evaluation criterion based on gradient-weighted summation of region-specific illumination and updates exposure parameters according to their first-order derivatives. Additionally, our method dynamically modulates exposure-update step sizes based on scene irradiance, ensuring rapid convergence under extreme lighting transitions. Integration with VO outputs further refines camera exposure allocation by determining appropriate shutter time and gain settings based on scene motion. Extensive experiments confirm that our approach effectively mitigates motion blur, rapidly responds to illumination changes, reduces image noise during stationary scenes, and enhances the overall robustness and real-time performance of contemporary visual SLAM algorithms.

CRediT authorship contribution statement

Qirui Sun: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation. **Qi Ouyang:** Methodology, Funding acquisition. **Xiaolong Li:** Visualization. **Zhaoying Cheng:** Visualization, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the Central Fund for Basic Scientific Research at Central Universities Defense Project (Grant No. 2024CDJGF-053). The authors would like to thank the Robotics Laboratory at Chongqing University for technical support. We also thank Prof. Qi Ouyang for his insights in algorithm optimization and Mr. Xiaolong Li for his experimental validation support.

Data availability

Data will be made available on request.

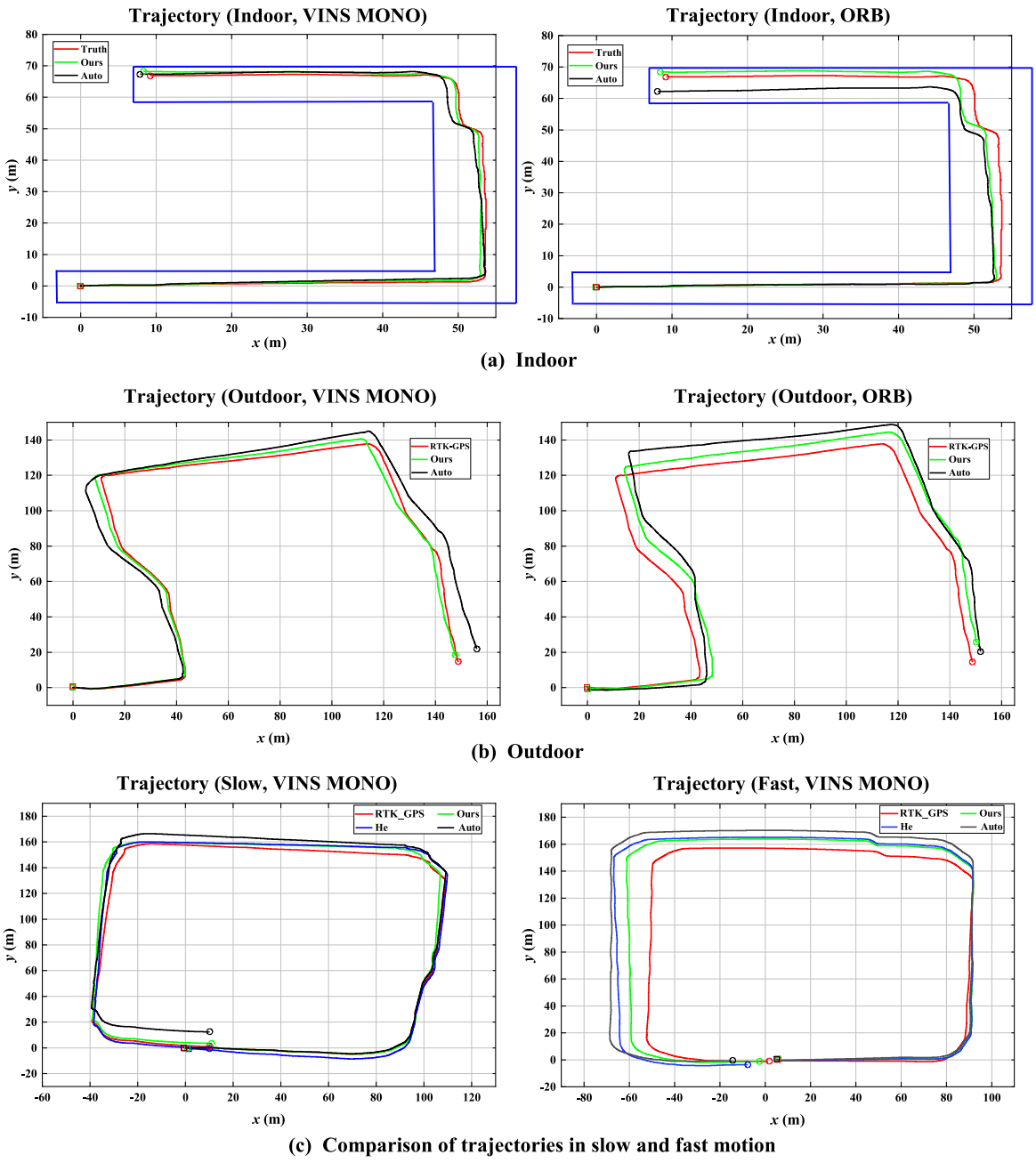


Fig. 10. Trajectory results of VINS-MONO and ORB-SLAM under different exposure control methods in indoor and outdoor scenarios.

Table 5
Processing time analysis of proposed method.

Methods	Processing time (ms) [640×480]			
Gradient-Weighted and Motion-Aware Camera Exposure Control	Region number	Image metric computation	Derivative computation	Total time
	200	6.22	7.37	8.42
	300	6.43	7.64	9.13
	400	6.56	8.42	9.57

References

[1] B.S. Zhang, X.G. Ma, H.J. Ma, C.B. Luo, DynPL-SVO: A robust stereo visual odometry for dynamic scenes, IEEE Trans. Instrum. Meas. 73 (2024) 1–10.

[2] T. Qin, P.L. Li, S.J. Shen, VINS-Mono: A robust and versatile monocular visual-inertial state estimator, IEEE Trans. Robot. 34 (4) (2018) 1004–1020.

[3] R. Mur-Artal, J.M.M. Montiel, J.D. Tardós, ORB-SLAM: A versatile and accurate monocular SLAM system, IEEE Trans. Robot. 31 (5) (2015) 1147–1163.

[4] J. Engel, V. Koltun, D. Cremers, Direct sparse odometry, IEEE Trans. Pattern Anal. Mach. Intell. 40 (3) (2018) 611–625.

[5] Y. Liu, Z. Ni, S. Wang, H. Wang, S. Kwong, High dynamic range image quality assessment based on frequency disparity, circuits and systems for video technology, IEEE Trans. (T-CSVT) 33 (8) (2023) 6.

[6] R. Gomez-Ojeda, Z.C. Zhang, J. Gonzalez-Jimenez, D. Scaramuzza, Ieee. Learning-based image enhancement for visual odometry in challenging HDR environments, in: IEEE International Conference on Robotics and Automation, ICRA, IEEE, 2018,

pp. 805–811.
[7] J. Tao, Y. Li, B.L. Guan, Y. Shang, Q.F. Yu, Simultaneous enhancement and noise suppression under complex illumination conditions, IEEE Trans. Instrum. Meas. 73 (2024).
[8] L. Taihang, G. Banglei, L. Minzu, L. Xiangyu, L. Jianbing, T. Jing, et al., Event-based multi-view photogrammetry for high-dynamic, high-velocity target measurement, Acta Mech. Sin. (2025).
[9] A. Agarwal, J. Hansrani, S. Bagwell, O. Rytov, V.R. Shah, K.L. Ong, et al., A 316MP, 120FPS, high dynamic range CMOS image sensor for next generation immersive displays, Sensors 23 (20) (2023) 8383.
[10] I. Shim, J.Y. Lee, I.S. Kweon, Auto-adjusting camera exposure for outdoor robotics using gradient information, in: IEEE/RSJ International Conference on Intelligent Robots and Systems, IROS, IEEE, 2014, pp. 1011–1017.
[11] I. Shim, T.H. Oh, J.Y. Lee, J. Choi, D.G. Choi, I.S. Kweon, Gradient-based camera exposure control for outdoor mobile platforms, IEEE Trans. Circuits Syst. Video Technol. 29 (6) (2019) 1569–1583.
[12] B. Han, Y.C. Lin, Y. Dong, H. Wang, T. Zhang, C.Y. Liang, Camera attributes control for visual odometry with motion blur awareness, IEEE ASME Trans. Mechatron. 28 (4) (2023) 2225–2235.
[13] K. Xu, Y.F. Hao, S.H. Yuan, C. Wang, L.H. Xie, AirSLAM: An efficient and illumination-robust point-line visual SLAM system, IEEE Trans. Robot. 41 (2025) 1673–1692.
[14] S.H. Cheng, C.H. Sun, S.J. Zhang, D.F. Zhang, SG-SLAM: A real-time RGB-D visual SLAM toward dynamic scenes with semantic and geometric information, IEEE Trans. Instrum. Meas. 72 (2023) 1–12.
[15] S. Kannapiran, N. Bendapudi, M.Y. Yu, D. Parikh, S. Berman, A. Vora, et al., Stereo visual odometry with deep learning-based point and line feature matching using an attention graph neural network, in: IEEE/RSJ International Conference on Intelligent Robots and Systems, IROS, IEEE, 2023, pp. 3491–3498.
[16] J. Tomasi, B. Wagstaff, S.L. Waslander, J. Kelly, Learned camera gain and exposure control for improved visual feature detection and matching, IEEE Robot. Autom. Lett. 6 (2) (2021) 2028–2035.
[17] S.Y. Zhang, J.H. He, Y.L. Zhu, J. Wu, J. Yuan, Efficient camera exposure control for visual odometry via deep reinforcement learning, IEEE Robot. Autom. Lett. 10 (2) (2025) 1609–1616.
[18] E. Rosten, T. Drummond, Machine learning for high-speed corner detection, in: European Conference on Computer Vision, Springer Berlin Heidelberg, 2006, pp. 430–443.
[19] E. Rublee, V. Rabaud, K. Konolige, G. Bradski, Ieee. ORB: an efficient alternative to SIFT or SURF, in: IEEE International Conference on Computer Vision, ICCV, IEEE, 2011, pp. 2564–2571.
[20] Z. Zhang, C. Forster, D. Scaramuzza, Active exposure control for robust visual odometry in HDR environments, in: 2017 IEEE International Conference on Robotics and Automation, ICRA, IEEE, 2017, pp. 3894–3901.
[21] J. Kim, Y. Cho, A. Kim, Ieee. Exposure control using Bayesian optimization based on entropy weighted image gradient, in: IEEE International Conference on Robotics and Automation, ICRA, IEEE, 2018, pp. 857–864.
[22] H. Lu, H. Zhang, S. Yang, Z. Zheng, Camera parameters auto-adjusting technique for robust robot vision, in: IEEE International Conference on Robotics and Automation, ICRA, IEEE, 2010, pp. 1518–1523.
[23] U. Shin, J. Park, G. Shim, F. Rameau, I.S. Kweon, Ieee. Camera exposure control for robust robot vision with noise-aware image quality assessment, in: IEEE/RSJ International Conference on Intelligent Robots and Systems, IROS, IEEE, 2019, pp. 1165–1172.
[24] Y.H. Li, L. Jin, M. Liu, Y.T. Mo, W.G. Zheng, D.Y. Ge, Y.D. Bai, Research on adaptive edge detection method of part images using selective processing, Process. 12 (10) (2024).
[25] Q.Z. Lan, Q. Tian, I.C. Soc, Gradient-guided knowledge distillation for object detectors, in: IEEE/CVF Winter Conference on Applications of Computer Vision, WACV, IEEE, 2024, pp. 423–432.
[26] T.X. Shan, B. Englot, D. Meyers, W. Wang, C. Ratti, D. Rus, Ieee. LIO-SAM: Tightly-coupled lidar inertial odometry via smoothing and mapping, in: IEEE/RSJ International Conference on Intelligent Robots and Systems, IROS, IEEE, 2020, pp. 5135–5142.



Qirui Sun received his Bachelor's degree in Intelligent Manufacturing Engineering from Soochow University, Suzhou, China, in 2023. He's currently pursuing a Master's degree in Control Science and Engineering at Chongqing University, Chongqing, China. His research interests include autonomous driving, multi-sensor fusion SLAM, and path tracking and control.



Qi Ouyang received the Ph.D. degree in Mechatronic Engineering from Chongqing University, China, in 2007. From 2011 to 2012, he was a visiting scholar at the University of Cambridge, UK. He is currently a Professor and Ph.D. supervisor with the School of Mechanical Engineering, Chongqing University. His research interests include autonomous driving perception technology, SLAM and navigation, as well as robotics actuation and control.



Xiaolong Li, male, originally from Zhoukou, Henan Province, China, is currently pursuing graduate studies at the School of Automation, Chongqing University. He received his Bachelor of Engineering degree from Chongqing University in 2019. His research primarily focuses on vehicle motion control and autonomous driving integration.



Zhaoying Cheng received her Bachelor's degree in Automation from Harbin University of Science and Technology, Harbin, China, in 2023. She is currently pursuing a Master's degree in Electronic Information at Chongqing University, China. Her research interests include machine vision, robotic path planning, robotic control, and multi-axis robotic control.